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Thermo-Elastic Model of Epicenter Displacement by Laser Pulse Irradiated on Metallic Surfaces

In recent years, there is a much interest in developing of nondestructive testing (NDT) systems using the pulse-echo laser ultrasonics. The key idea is to combine a low-power and short-pulsewidth laser excitation with a continuous sensing laser; and use a scanning mechanism, such as five degrees-of-freedom (5DOF)-axis robot, laser mirror scanner, or motorized linear translation or rotation scanner stage, to scan the combined beam on the structure. In order to optimize the parameters of the excitation laser, a realistic theoretical model of the epicenter displacement in thermo-elastic regime is needed. This paper revisits and revises the study of Spicer and Hurley (1996, "Epicentral and Near Epicenter Surface Displacements on Pulsed Laser Irradiated Metallic Surfaces," Appl. Phys. Lett., 68(25), pp. 3561–3563) on thermo-elastic model of epicenter displacement with two new contributions: first, we revised Spicer's model to take into account the optical penetration effect, which was neglected in Spicer's model; and second, the revised model was used to investigate the effect of laser rise time and beam size to the epicenter displacement. We showed that a pulse laser with short rise time generates an equivalent surface displacement with a pulse laser with long rise time, except a "spike" at the beginning of the epicenter waveform; also when the laser beam size increases, the epicenter displacement decreases. These two conclusions were then validated by experiments. [DOI: 10.1115/1.4038030]

Keywords: laser thermo-elastic model, epicenter displacement, nondestructive testing, inverse Hankel–Laplace transform

1 Introduction

Laser ultrasonics techniques for nondestructive testing (NDT) of aeronautic industry have been developed for several decades. Initial studies in the 1970s and 1980s [1–3] showed strong potential for industrial applications. In those studies, the pulse-echo mode of laser ultrasonics was more favorable than through-transmission mode, since it requires the access of only one surface of the structure. In the 1990s, several groups investigated the use of laser NDT for composite structures [4–7]. Today, there are several companies provide commercial laser NDT such as PaR Systems [8], Tecnar Automation [9], iPhoton Solutions [10], and Tecnom [11]. In academic, this topic is also attracted much interested with several laboratory-scale systems [12–14]. A mobile pulse-echo ultrasonic propagation imaging system based on linear translation scanning of laser beams has been developed and delivered to Korea Air Force in 2016, and the system is now in-service [15].

The key idea of the pulse-echo laser NDT systems [8–15] is laser-sensing echo mode of ultrasound generated by a short pulse laser on the same surface; and use a scanning mechanism to scan the sensing and excitation laser beams, such as five degrees-of-freedom (5DOF)-axis robot [12], laser mirror scanner [8–11], and motorized translation or rotation scanner stage [13–15]. Each scanning mechanism has its own advantages as well as limitations. The use of a 5DOF-axis robot for scanning allows inspection of very complex shaped objects such as small radius of curvature structures; however, the scan speed is very slow with

the max scan speed in Ref. [12] is 30 Hz. Laser mirror scanner provides extremely fast scan speed (up to kHz range) and has compact size; on the other hand, in case of large scan area, the angle of incidence of the sensing beam is significant near the edges of the scan area, which causes the absolute level of light intensity reflected back from the structure surface reduced significantly; as the result, the signal quality is nonuniform across the scan structure, which is problematic for analysis. In the case the structure is flat or mildly contoured, the linear translation scanner stage avoids the issue with high angle scan, though it is much bulkier.

A realistic modeling of the laser ultrasonic waves in thermo-elastic regime is essential for optimization of laser parameters for the laser NDT systems [16]. Early studies on this topic focused on developing analytical solutions with the aid of inverse transformation such as Hankel and Laplace transform. In those theoretical studies, the models were simplified so that a solution can be obtained. Scruby et al. [17] proposed the point-source representation where the laser heated area is equivalent to a set of two mutually orthogonal force dipoles. Rose [18] provided a rigorous mathematical treatment of the point-source representation and called it the surface center of expansion (SCOE) model. Though SCOE correctly predicts the major features of laser ultrasonic waveform, it fails to predict the precursor, which is a small positive spike at the beginning of the epicenter waveform [19]; it is because SCOE neglects the thermal diffusion from the heat source. Spicer and Hurley [20] developed a thermo-elastic model including thermal diffusion to calculate the surface displacement of a metallic surface under the irradiation of a laser pulse. More realistic model with consideration of optical penetration effect was studied by Arias and Achenbach [21]; however, this study used the laser line-source representation instead of circular-source

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representation, which is more commonly used in laser NDT systems. When the model becomes too complicated, for example, temperature-dependent parameters as well as complex geometry, finite element method (FEM) can be used [22–24]. Wang et al. [24] investigated the generation of the elastic waves in nonmetallic materials in the thermo-elastic regime using FEM, taking into account a full model with thermal diffusion, finite spatial and temporal shape of the laser pulse, optical penetration, and temperature dependence of material properties.

A majority of existing studies of both analytical and FEM solutions deal with the surface waves and the through-transmission waves of laser ultrasonics. The main reason is due to the widespread use of those wave modes in NDT. In pulse-echo mode, the setup is more complicated, since optical sensing must be used to avoid the obstruction of the laser excitation beam; also often an optical assembly must be used to combine the sensing and excitation laser beams. Spicer's model [20] deals directly with the problem of epicenter displacement in pulse-echo mode; however, this pioneer paper did not consider optical penetration effect. This paper revisits and revises the study of Spicer and Hurley [20] with two new contributions: first, we revised Spicer's model to take into account the optical penetration effect, which was neglected in Spicer's model; and second, the revised model was used to investigate the effect of laser rise time and beam size to the epicenter displacement. This paper is organized as follows: Section 2 shows the formulation and numerical study of the thermo-elastic model of epicenter displacement. Experimental data to support the theoretical study are showed in Sec. 3. Section 4 is conclusion and discussion.

2 Formulation and Numerical Study of the Thermo-Elastic Model of Epicenter Displacement

2.1 Formulation. Figure 1(a) shows the schematic of the epicenter displacement produced by a short pulse laser. The thermo-elastic field is governed by the coupled equations of thermo-elasticity [20,21,25]

$$\nabla^2 T - \kappa^{-1} \dot{T} - c_H^{-2} \ddot{T} = -k^{-1} W \quad (1)$$

$$(\lambda + 2\mu) \nabla \varphi - 2\mu \nabla \times \psi = \rho \ddot{u} + \gamma \nabla T \quad (2)$$

where κ is thermal diffusivity, $c_H = \sqrt{\kappa/\tau}$ is heat wave speed, τ is material relaxation time, k is thermal conductivity, W is heat source power density, λ and μ are Lamé coefficients, φ and ψ are dilatational and rotational potentials, ρ is density, $\gamma = (3\lambda + 2\mu)\alpha_T/(\lambda + 2\mu)$ is thermo-acoustic coupling constant, α_T is volumetric thermal expansion coefficient, T is temperature rise

above ambient, ∇ is gradient, $\nabla \times$ is curl, and ∇^2 is Laplace operator defined in cylindrical coordinates (r, θ, z) .

The displacement vector u can be expressed as

$$u = \nabla \varphi + \nabla \times \psi \quad (3)$$

where the potentials satisfy the equations

$$\nabla^2 \varphi - a^2 \ddot{\varphi} = \gamma T \quad (4)$$

$$\nabla^2 \psi - b^2 \ddot{\psi} = 0 \quad (5)$$

where $a = c_1^{-1}$ and $b = c_2^{-1}$, c_1 and c_2 are longitudinal and transverse speed, respectively.

The heat source power density W can be expressed as [21]

$$W = E(1 - R_i)h(z)f(r)g(t) \quad (6)$$

with the temporal profile $g(t)$, spatial profile $f(r)$, and absorption profile $h(z)$ are [26,27]

$$h(z) = \gamma_e e^{-\gamma_e z} \quad (7)$$

$$f(r) = \frac{1}{\sqrt{2\pi}} \frac{2}{R_G} e^{-2r^2/R_G^2} \quad (8)$$

$$g(t) = \frac{t}{v^2} e^{-t/v} \quad (9)$$

where E is pulse energy, R_i is surface reflectivity, γ_e is extinction coefficient, v is laser pulse rise time, and R_G is $1/e^2$ half-width of the Gaussian beam. It is noted that the assumed temporal profile $g(t)$ in Eq. (9) gives the laser pulsewidth $T \approx 2.5 v$ [28]. The profiles $g(t)$, $f(r)$, and $h(z)$ are shown in Fig. 1(b).

The boundary conditions at the surface, $z = 0$

$$\partial T / \partial z|_{z=0} = 0 \quad (10)$$

$$\sigma_{zz}|_{z=0} = -\gamma T + \lambda \nabla^2 \varphi + 2\mu \frac{\partial}{\partial z} \left(\frac{\partial \varphi}{\partial z} + \frac{1}{r} \frac{\partial}{\partial r} (r\psi) \right) \Big|_{z=0} = 0 \quad (11)$$

$$\sigma_{zr}|_{z=0} = \mu \left[\frac{\partial}{\partial z} \left(\frac{\partial \varphi}{\partial r} - \frac{\partial \psi}{\partial r} \right) + \frac{\partial}{\partial r} \left(\frac{\partial \varphi}{\partial z} + \frac{1}{r} \frac{\partial}{\partial r} (r\psi) \right) \right] \Big|_{z=0} = 0 \quad (12)$$

Transformation technique is used to solve the system of Eqs. (1)–(12). Hankel transform is applied to spatial domain, and

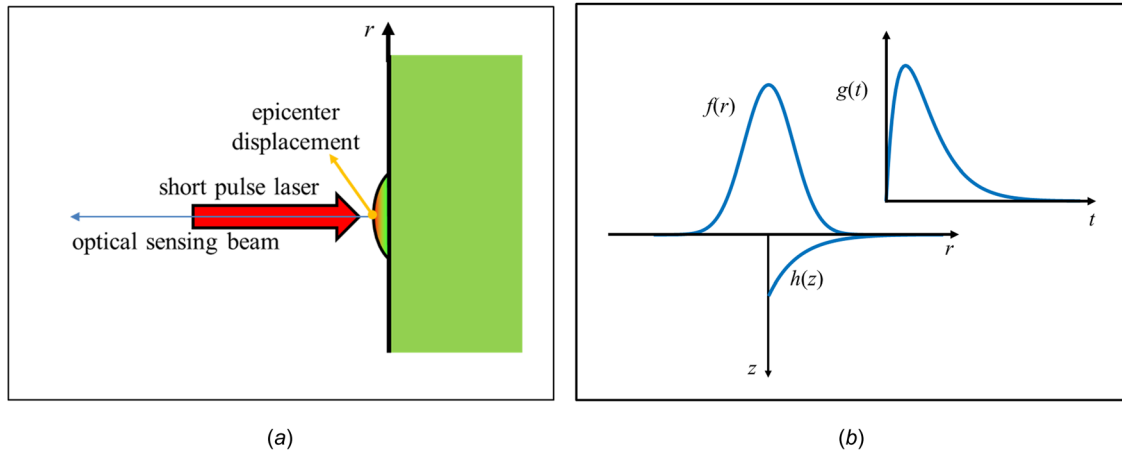


Fig. 1 (a) Epicenter displacement produced by a short pulse laser and measured using an optical sensing beam. (b) Temporal profile $g(t)$, spatial profile $f(r)$, and absorption profile $h(z)$ of the laser pulse.

Table 1 Material parameters used in the numerical study

Parameters	Notation	Unit	Aluminum	Carbon fiber reinforced polymer (CFRP) composite
Surface reflectivity	R_i	—	0.90	0.75
Thermal expansion coefficient	α_T	K^{-1}	2.3×10^{-4}	1×10^{-6}
Thermal diffusivity	κ	$m^2 s^{-1}$	1×10^{-4}	1×10^{-6}
Thermal conductivity	k	$W m^{-1} K^{-1}$	240	0.2
Longitudinal velocity	c_1	$m s^{-1}$	6321	3000
Transverse velocity	c_2	$m s^{-1}$	3110	2000
Thermo-acoustic coupling	γ	K^{-1}	4.48×10^{-5}	1.2×10^{-6}
Extinction coefficient	γ_e	m^{-1}	2×10^8	0.7×10^7

Laplace transform is applied to time domain. Hankel and Laplace transforms are defined as [25]

$$W_p(p) = \mathcal{H}[f(r)] = \int_0^\infty r f(r) J_0(pr) dr \quad (13)$$

$$W_s(s) = \mathcal{L}[g(t)] = \int_0^\infty e^{-st} g(t) dt \quad (14)$$

and their inverse transforms are

$$f(r) = \mathcal{H}^{-1}[W_p(p)] = \int_0^\infty p W_p(p) J_0(pr) dp \quad (15)$$

$$g(t) = \mathcal{L}^{-1}[W_s(s)] = \frac{1}{2\pi i} \int_{\sigma_p - i\infty}^{\sigma_p + i\infty} e^{st} g(t) dt \quad (16)$$

where J_0 is the zero-order Bessel function of the first kind, σ_p is greater than the real part of all singularities of $W_s(s)$, (t, s) is the Laplace pair, and (p, r) is the Hankel pair.

Assume that heat wave speed equals longitudinal wave speed [19], the thermo-elastic problem in the transformed domain can be derived as

$$\left(p^2 + a^2 s^2 + \frac{s}{\kappa}\right) \tilde{T} - \frac{\partial^2 \tilde{T}}{\partial z^2} = \frac{\tilde{W}}{k} \quad (17)$$

$$(p^2 + a^2 s^2) \tilde{\phi} - \frac{\partial^2 \tilde{\phi}}{\partial z^2} = -\gamma \tilde{T} \quad (18)$$

$$(p^2 + b^2 s^2) \tilde{\psi} - \frac{\partial^2 \tilde{\psi}}{\partial z^2} = 0 \quad (19)$$

Solution of out-of-plane displacement at the surface $z=0$ in transform domain can be derived as [21]

$$\begin{aligned} \tilde{u}_z|_{z=0} &= \gamma b^2 k^{-1} s^2 (\beta^2 + p^2) \\ &\times \left[\frac{\kappa}{s} (\alpha \xi^{-1} - 1) + \frac{1}{\xi^2 - \gamma_e^2} (\alpha \gamma_e^{-1} - 1) \right] \frac{\gamma_e^2}{\gamma_e^2 - \xi^2} R^{-1} \tilde{W} \end{aligned} \quad (20)$$

where $\alpha^2 = a^2 s^2 + p^2$, $\beta^2 = b^2 s^2 + p^2$, $\xi^2 = \alpha^2 + s/\kappa$, $R = (\beta^2 + p^2)^2 - 4\alpha\beta p^2$, $\tilde{W} = W_0 W_p(p) W_s(s)$, $W_0 = E(1 - R_i)$, $W_p(p) = (1/\sqrt{2\pi}) e^{-p^2 R_G^2/8}$, and $W_s(s) = \mathcal{L}((t/\nu^2) e^{-t/\nu}) = 1/(\nu s + 1)^2$.

Note that if assuming no optical penetration ($\gamma_e \rightarrow \infty$), then Eq. (20) is the same as derived in Ref. [20]

$$\{\tilde{u}_z|_{z=0}\}_{\gamma_e \rightarrow \infty} = \gamma \kappa b^2 k^{-1} [s(\beta^2 + p^2)(\alpha \xi^{-1} - 1) R^{-1}] \tilde{W} \quad (21)$$

Finally, the out-of-plane displacement at the surface $z=0$ is calculated by inverse Hankel–Laplace transform

$$u_z|_{z=0} = \frac{1}{2\pi i} \int_0^\infty \int_{\sigma_p - i\infty}^{\sigma_p + i\infty} p J_0(pr) e^{st} \tilde{u}_z|_{z=0} ds dp \quad (22)$$

The solution of surface displacement in Eq. (22) can be solved using numerical method. Weeks method [29] was used for Laplace inverse transform, and global adaptive quadrature method [30] was used for Hankel inverse transform.

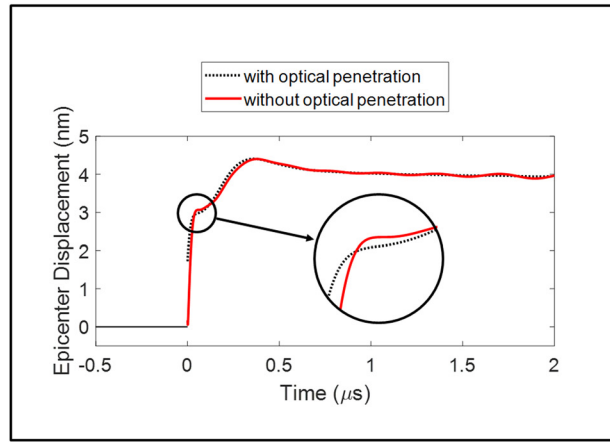
2.2 Numerical Study. In this section, we present the numerical study with the thermo-elastic model described in Sec. 2.1 to investigate the effect of several parameters to the epicenter displacement. Section 2.2.1 studies the effect of optical penetration with thermo-elastic parameters of aluminum [21] and CFRP composite materials [31,32] as shown in Table 1. Sections 2.2.2 and 2.2.3 study the effect of laser pulse rise time and laser beam size with the thermo-elastic parameters of aluminum material. The values used for the laser parameters are laser pulse energy $E = 1 \times 10^{-3}$ J, laser pulse rise time $\nu = 8 \times 10^{-9}$ s, and laser beam radius $R_G = 1 \times 10^{-3}$ m.

2.2.1 Effect of Optical Penetration Parameter. Figure 2(a) shows the comparison of the epicenter displacement with and without consideration of optical penetration in the case of aluminum material. To make sure that readers can realize there is indeed a slight difference between two cases, a zoom of a portion of two curves is showed. The waveform of the epicenter displacement can be explained as follows. At the starting time point $t=0$, the short pulse laser heats the surface of the specimen and causes a quick local thermal expansion at the center of the heat source. The near field surface waves from lateral extent of the laser beam come later, which causes a peak at $0.38 \mu s$. Increasing the beam diameter delays the arrival waves, and as the results, shifts the peak of the epicenter waveform to the right, as shown later in Sec. 2.2.3. Similarly, increasing the laser rise time spreads the energy deposition in time and shifts the peak of the epicenter waveform to the right, as shown later in Sec. 2.2.2.

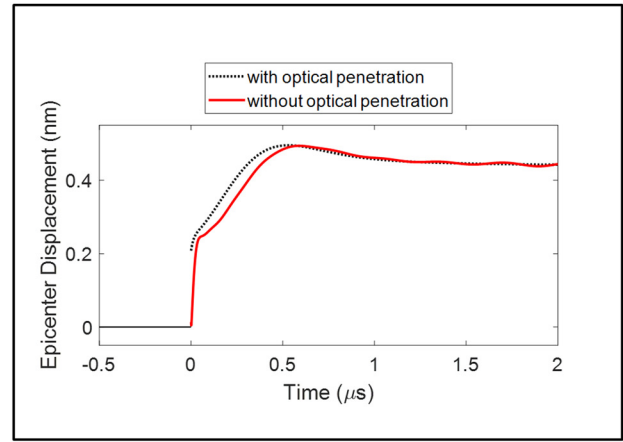
The relative difference of epicenter displacement with and without consideration of optical penetration in Fig. 2(a) with aluminum material is approximately 0.82%. This result shows that the assumption of no optical penetration is reasonable with metallic materials [20]. It is readily understood since the thermal diffusion length for aluminum $\sqrt{4\kappa T} \approx 2.8 \times 10^{-6}$ (m) is much larger than the inverse of extinction coefficient 0.5×10^{-8} (m), which means that the optical penetration effect must be small compared with thermal diffusion effect and can be neglected.

The effect of optical penetration is more significant in the case of composite material, as shown in Fig. 2(b). It is because the thermal diffusion length in the case of composite is $\sqrt{4\kappa T} \approx 2.8 \times 10^{-7}$ (m) in the same order of the inverse of extinction coefficient $\approx 1.4 \times 10^{-7}$ (m). The amplitude of the epicenter waveform in the first 500 ns increases when the model includes optical penetration effect. However, optical penetration has little effect to the tail of the epicenter displacement waveform.

2.2.2 Effect of Laser Pulse Rise Time. A parametric study is carried out to investigate the influence of the laser pulse rise time



(a)



(b)

Fig. 2 Comparison of epicenter displacement with and without consideration of optical penetration: (a) aluminum material and (b) CFRP composite material

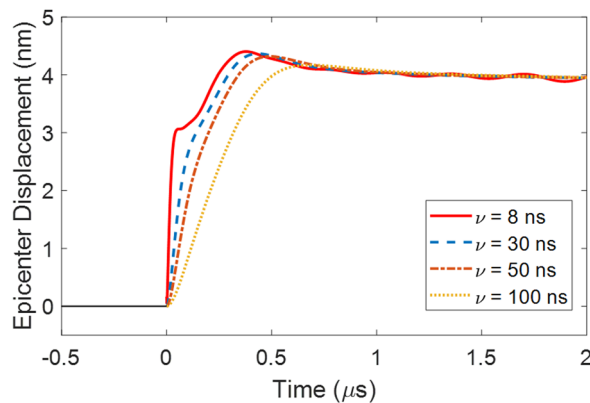


Fig. 3 Epicenter displacement produced by laser pulse irradiated on aluminum surface with different rise time of pulse laser

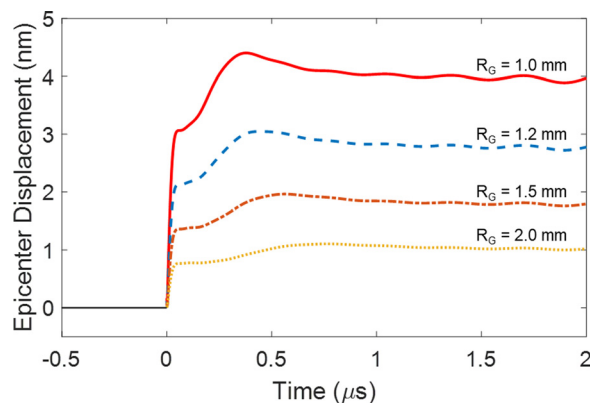


Fig. 4 Epicenter displacement produced by laser pulse irradiated on aluminum surface with different beam size

on the epicenter displacement produced by laser pulse irradiated on aluminum surface. The laser pulse rise time (ν) is varied from 8 ns to 100 ns with a fixed value of beam energy ($E = 1$ mJ) and beam radius ($R_G = 1$ mm). The calculated time signals are shown in Fig. 3. Apart from the difference at the beginning of the waveform (up to about 500 ns), the epicenter displacement after ≈ 1 μ s (the “tail” of the waveform) is insensitive to the change of laser pulse rise time.

Figure 3 shows that, in effect, the laser pulse raises the surface of the irradiating area with the magnitude of 4 nm as shown in the tail of the waveform. In a homogeneous, isotropic, linear elastic medium, the propagating waves after that are proportional to amplitude of the tail.

2.2.3 Effect of Laser Beam Size. A parametric study is carried out to investigate the influence of the laser beam size on the epicenter waveform produced by laser pulse irradiated on aluminum surface. The laser beam radius (R_G) is varied from 1 mm to 2 mm with a fixed value of beam energy ($E = 1$ mJ) and laser pulse rise time ($\nu = 8$ ns). The calculated time signals are shown in Fig. 4. It is shown that as the laser beam size increases, the amplitude of the epicenter displacement decreases. Also the signals area broader, and the peaks move the right of the time axis.

3 Experiment

3.1 Experiment Setup. Figure 5 shows the setup of the experiment. A Nd:YAG laser (1064 nm, 1 mJ) was used for generation of the laser pulse. The surface of the aluminum specimen after experiment has no indication of thermal damage; therefore, the laser ultrasonic waves were generated in thermal-elastic regime. The laser pulsewidth of this laser can be varied from 20 ns to 50 ns, which gives the laser pulse rise time varied from 8 ns to 20 ns. The sensing device was a Polytec OFV-505 with the chosen setting of 1 MHz bandwidth. An aluminum specimen was used with the thickness of 4 mm; the surface of the aluminum was treated so that it is clean and highly reflective. The measured epicenter ultrasonic signals then went through a programmable amplifier-filter and recorded using an oscilloscope. An optical assembly was designed to combine the pulse laser beam and the continuous sensing beam. To make sure the beam size was measured precisely, we used a beam profiler putting at the location of the aluminum specimen in Fig. 5. Examples of profiler results with the measured beam diameters of 2 mm and 4 mm, which corresponding to beam radius $R_G = 1$ mm and $R_G = 2$ mm, are shown in Figs. 6(a) and 6(b).

Since the sensing device used in this experiment has very low sampling frequency, it is not possible to obtain the exact theoretical waveforms in Figs. 2–4 (in the paper of Spicer and Hurley [20], a sensor with bandwidth of 80 MHz was used). However, we will show that the peak-to-peak amplitude of the experiment waveform can be used to verify two conclusions in Sec. 2: first, the amplitude of tail of the epicenter waveform is insensitive with the change of laser rise time, and second, when the laser beam

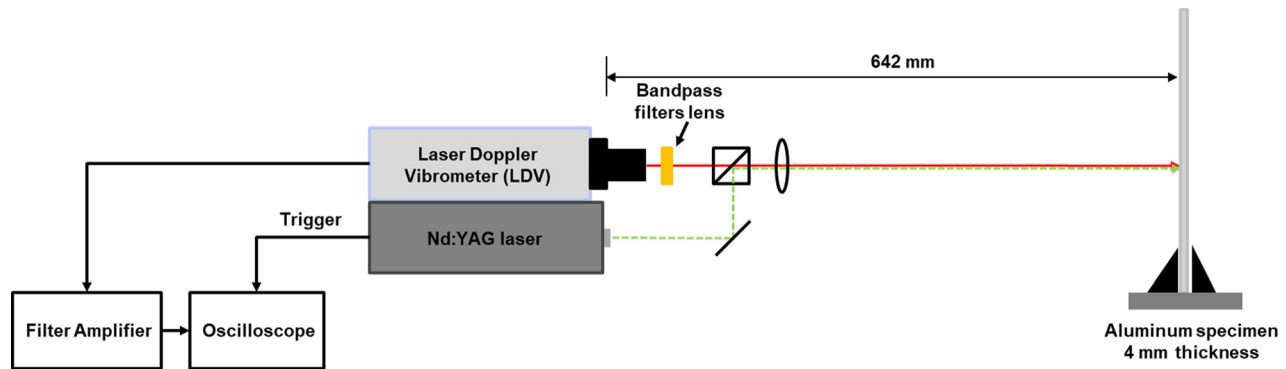


Fig. 5 Experiment setup to measure the epicenter pulse-echo laser ultrasonics

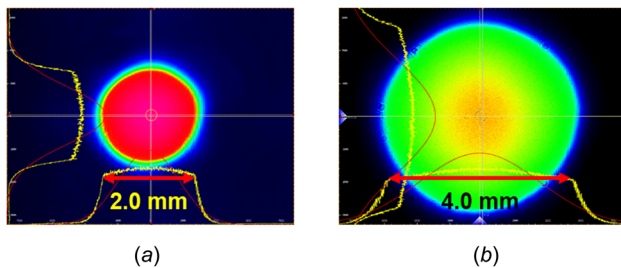


Fig. 6 (a) Beam profile result of 2-mm diameter laser beam and (b) beam profile result of 4-mm diameter laser beam

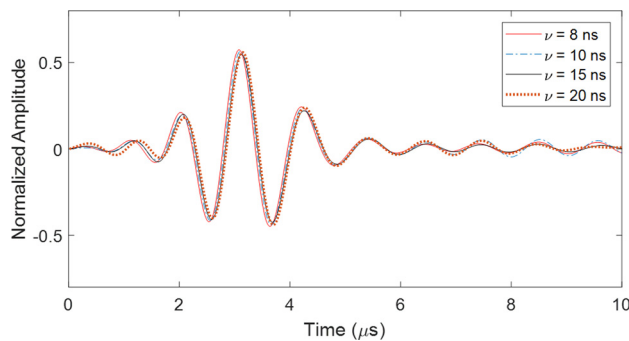


Fig. 7 Experimental epicenter waveform with varied laser rise time

size increases, the epicenter displacement decreases at a constant beam energy.

3.2 Experimental Results. Figure 7 shows the experimental epicenter waveforms with varied laser rise time ν (from 8 ns to 20 ns), constant beam energy ($E = 1$ mJ), and constant beam radius ($R_G = 1$ mm). The signals in Fig. 7 are normalized by dividing to the peak-to-peak amplitude of the waveform $\nu = 20$ ns. It is clear that the change of laser rise time has little effect to the peak-to-peak amplitude of the waveform as predicted in Sec. 2.2.2. Figure 8 shows the experimental epicenter waveforms with varied laser beam radius R_G (from 1 mm to 2 mm), constant beam energy ($E = 1$ mJ), and constant laser pulse rise time ($\nu = 8$ ns). The signals in Fig. 8 are normalized by dividing to the peak-to-peak amplitude of the waveform $R_G = 2$ mm. The amplitude of the signals decreases when the laser beam radius increases as predicted in Sec. 2.2.3.

The ratios of amplitude changes when changing laser rise time and laser beam radius are shown in Figs. 9 and 10, respectively. A well-agreement between theoretical and experimental results is

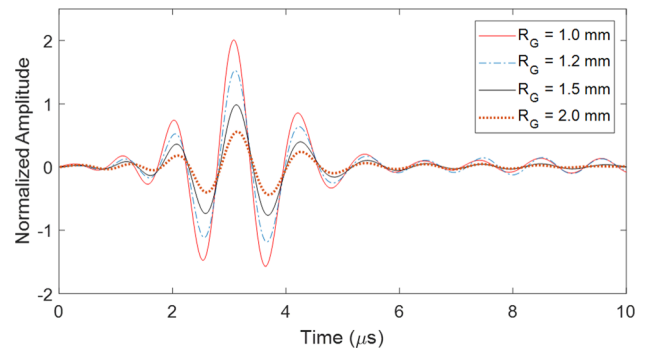


Fig. 8 Experimental epicenter waveform with varied laser beam radius with a constant energy

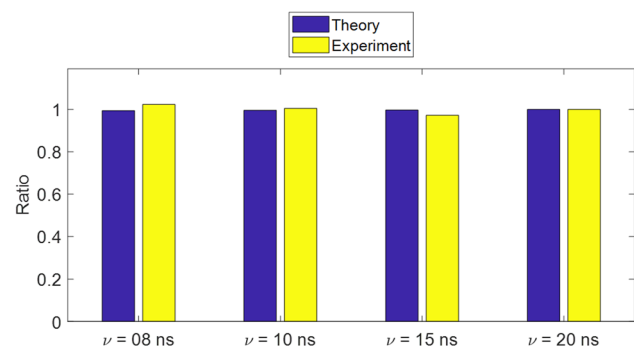


Fig. 9 Peak-to-peak amplitude of the normalized experimental waveform versus normalized amplitude of theoretical epicenter displacement with varied laser rise time

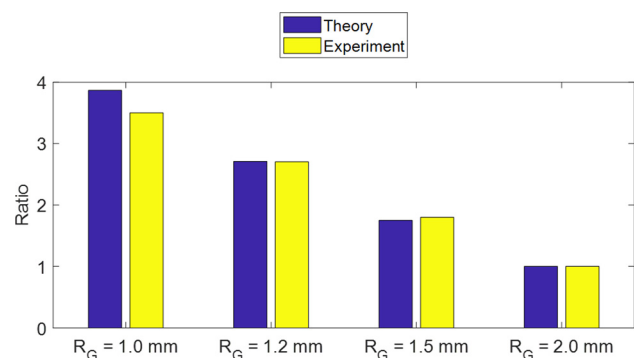


Fig. 10 Comparison ratios of amplitude changes when changing laser beam radius between theory and experiment

found. Figure 9 shows that the laser rise time has little effect to the peak-to-peak amplitude of the experimental waveform as well as theoretical epicenter displacement; and Fig. 10 shows that the peak-to-peak amplitude of the experimental waveform and theoretical epicenter displacement decreases as laser beam radius increases. For example, when the laser radius reduces from 2 mm to 1 mm, the peak-to-peak amplitude of the experimental waveform increase 3.5 times, whereas the amplitude of theoretical epicenter displacement increases 3.9 times.

4 Conclusions and Discussion

In this paper, a thermo-elastic model of the epicenter displacement was presented. This model takes into account the effects of thermal diffusion, optical penetration, and the information of spatial and temporal distribution of a circular laser source. The thermo-elastic problem was solved using the Hankel–Laplace transformation technique. The inversion of the Hankel–Laplace transform was calculated numerically to obtain theoretical epicenter waveforms. It is found that the contribution of the optical penetration effect in aluminum material is small and therefore can be neglected. On the other hand, the optical penetration effect is significant with composite material and must be included in the thermo-elastic model.

Using the developed numerical model, we further investigated the effect of laser rise time and beam size to the epicenter displacement. It is showed that a pulse laser with short rise time generates an equivalent surface displacement with a pulse laser with long rise time, except a spike at the beginning of the epicenter waveform; also when the laser beam size increases, the epicenter displacement decreases. These two conclusions were then validated by experiments.

This study of laser parameters is useful for the selection of the laser excitation of a laser NDT system. It is well-known that typically, the shorter of laser pulsewidth, the higher of the cost of the laser, given that the laser beam energy is the same. Since we concluded that a laser with pulsewidth of 40 ns can generate the equivalent laser ultrasound amplitude with a laser with pulsewidth of 8 ns, the former should be chosen for the NDT system. However, note that this conclusion is only valid in a certain range of laser pulsewidth. For example, this conclusion is not applicable for ultrashort pulse laser such as femtosecond (10^{-15} s) laser [33] or long pulse laser such as milliseconds (10^{-3} s) laser.

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